

Explanation-Supervised Attention for Multi-Label Thoracic Disease Classification: A Variant CNN Trained from Weak Localization Labels

Israel Agyekum Joel Dadi-Klutse Eric Okyere

University of Ghana — Department of Computer Science

CSCD 618 / DSCD 604 — Algorithmic Track

June 2026

0 Abstract

Chest radiography is the most accessible thoracic imaging modality, yet accurate interpretation depends on scarce radiological expertise. Convolutional neural networks achieve expert-level classification performance on the NIH ChestX-ray14 benchmark but function as opaque black boxes. The standard remedy—post-hoc Grad-CAM—is computed after training and is not optimised to be correct; its heat maps may be unfaithful and clinically misleading. We address this gap by proposing a variant CNN whose spatial attention module is a first-class trainable output supervised directly by clinician-annotated bounding-box masks, converting explanation from a post-hoc diagnostic into part of the optimisation objective. The model additionally incorporates a label-correlation-aware loss that penalises prediction patterns inconsistent with empirical clinical co-occurrence. We evaluate seven configurations—three backbone baselines (ResNet50, DenseNet121, EfficientNet-B0) and four attention variants—on both classification (macro ROC-AUC, F1) and localisation (IoU, pointing-game accuracy against ground-truth bounding boxes). Statistical significance is assessed via DeLong’s test. The central research question is whether explanation supervision improves localisation faithfulness while keeping AUC competitive at zero additional inference cost.

Keywords: explanation-supervised attention, thoracic disease classification, ChestX-ray14, CBAM, focal loss, label co-occurrence, Grad-CAM, localisation faithfulness.

0 Contents

Abstract	1
1 Introduction	3
1.1 Research Questions	3
1.2 Contributions	4
2 Related Work	4
2.1 CNN Backbones for Medical Imaging	4
2.2 Multi-Label Classification on ChestX-ray14	4
2.3 Attention Mechanisms	5
2.4 Post-Hoc Explanation Methods	5
2.5 Right-for-the-Right-Reasons Training	5
2.6 Label Dependency Modelling	6
3 Methodology	6
3.1 Dataset	6

3.2	Model Architecture	7
3.2.1	CNN Backbone	8
3.2.2	Channel Attention (Optional)	8
3.2.3	Supervised Spatial Attention Module	8
3.2.4	Classifier Head	8
3.3	Loss Function	8
3.3.1	Classification Loss ($\mathcal{L}_{\text{cls}}$)	9
3.3.2	Attention Loss ($\mathcal{L}_{\text{attn}}$)	9
3.3.3	Label-Correlation Loss ($\mathcal{L}_{\text{corr}}$)	9
3.4	Training Protocol	10
4	Experimental Setup	10
4.1	Model Configurations	10
4.2	Evaluation Metrics	11
4.2.1	Classification	11
4.2.2	Localisation	11
4.3	Compute	11
5	Results	12
5.1	Classification Performance	12
5.2	Localisation Performance	12
5.3	Ablation Study	13
5.4	Training Curves and Qualitative Results	14
6	Discussion	18
6.1	Classification vs Localisation Trade-off	18
6.2	Localisation Faithfulness	18
6.3	Ablation Insights	18
6.4	Backbone Comparison	19
6.5	Limitations	19
7	Conclusion	19
7.1	Future Work	20
A	Full Per-Class ROC-AUC	22

1 Introduction

Thoracic diseases—including pneumonia, pleural effusion, atelectasis, emphysema, and pneumothorax—are among the leading causes of global morbidity and mortality. Chest radiography (CXR) is the front-line and most widely accessible imaging modality, yet accurate interpretation requires experienced radiologists whose availability is unevenly distributed globally. This gap is particularly acute in resource-limited settings, including much of sub-Saharan Africa, where diagnostic delay and inter-reader inconsistency remain significant barriers to care.

Deep convolutional neural networks have demonstrated expert-level classification performance on CXR datasets, matching or exceeding radiologist accuracy on several findings. Despite impressive predictive performance, these models function as opaque black boxes: clinicians cannot verify whether a prediction is grounded in clinically relevant image regions, which limits trust and clinical deployment in safety-critical settings.

Post-hoc saliency methods are widely used to generate spatial explanations for CNN decisions. However, such explanations are computed after training on a fixed model and are not optimised to be correct—their heat maps frequently fail to align with radiologist-annotated lesion regions and can be misleading in clinical practice. [Sun et al. \[2023\]](#) showed that inherently interpretable classifiers can achieve competitive accuracy while producing explanations that more faithfully reflect clinically relevant regions. This motivates a fundamentally different approach: rather than explaining a fixed model after the fact, train the model to attend to the right regions from the outset.

We propose a variant CNN with an explanation-supervised spatial attention module. The key observation is that NIH ChestX-ray14 provides clinician-annotated bounding boxes for 8 of 14 findings across 880 images. Although this weak localisation signal is insufficient for full segmentation training, it is adequate to supervise the model’s spatial attention map through an alignment loss, converting explanation from a post-hoc artefact into part of the optimisation. The model also incorporates a label-correlation regulariser that penalises prediction vectors inconsistent with empirical clinical co-occurrence patterns.

1.1 Research Questions

This work addresses three central questions:

- RQ1.** Does explanation supervision measurably improve localisation faithfulness (IoU, pointing-game accuracy) relative to post-hoc Grad-CAM, while maintaining competitive macro AUC?
- RQ2.** What is the individual contribution of the attention loss ($\mathcal{L}_{\text{attn}}$) and the label-correlation loss ($\mathcal{L}_{\text{corr}}$) as isolated ablations?

RQ3. How does the choice of CNN backbone (ResNet50, DenseNet121, EfficientNet-B0) interact with the attention supervision mechanism?

1.2 Contributions

1. **Explanation-supervised attention module:** a CBAM-style spatial attention branch whose map $\mathbf{A}$ is a first-class trainable output supervised by bounding-box masks via a composite alignment loss (Dice + MSE on annotated images; L1 sparsity on unannotated images).
2. **Label-correlation-aware objective:** a lightweight co-occurrence regulariser appended to the training loss, requiring no architectural change at inference time.
3. **Systematic evaluation:** seven model configurations across classification and localisation metrics, with DeLong’s test for statistical significance.

2 Related Work

2.1 CNN Backbones for Medical Imaging

Modern CNN backbones have advanced significantly in both accuracy and efficiency. [Liu et al. \[2022\]](#) demonstrated that a purely convolutional architecture (ConvNeXt), modernised by borrowing design principles from vision transformers, matches transformer performance on ImageNet while retaining the inductive biases of convolutions that are beneficial for structured medical images such as chest radiographs. In parallel, [Liu et al. \[2021\]](#) proposed the Swin Transformer, a hierarchical vision transformer using shifted windows that achieves strong performance across recognition and dense prediction tasks, earning the ICCV 2021 best paper award. For efficient deployment, [Tan and Le \[2021\]](#) introduced EfficientNetV2, which trains up to $11\times$ faster than prior state-of-the-art models while using fewer parameters through training-aware neural architecture search. We evaluate ResNet50, DenseNet121, and EfficientNet-B0 as backbone baselines; these legacy architectures remain competitive on mid-scale medical imaging datasets and provide controlled ablation points, as confirmed by recent multi-architecture benchmarks on ChestX-ray14 [[Holste et al., 2022](#)].

2.2 Multi-Label Classification on ChestX-ray14

The NIH ChestX-ray14 dataset (112,120 images, 14 findings) remains the largest publicly available labelled chest radiograph collection and the de facto benchmark for multi-label thoracic disease classification. [Holste et al. \[2022\]](#) provided a rigorous re-evaluation of this benchmark under realistic long-tailed class distributions, demonstrating that standard training recipes are significantly biased towards frequent classes and proposing balanced

evaluation protocols that we adopt in our own experimental setup. The severe positive-negative label imbalance inherent to multi-label CXR classification motivates dedicated loss functions; [Ridnik et al. \[2021\]](#) proposed the Asymmetric Loss (ASL), which applies different focusing strengths to positive and negative samples, and achieved state-of-the-art results on multiple multi-label benchmarks including MS-COCO and Open Images. We adopt ASL-inspired weighting in our classification objective.

2.3 Attention Mechanisms

Spatial attention has become a central component of both CNN and transformer architectures for medical image analysis. [Khader et al. \[2025\]](#) recently proposed CLARiT_y, a vision transformer for multi-label CXR classification and weakly-supervised localisation that leverages self-attention to implicitly identify discriminative regions for each pathology. [Chefer et al. \[2021\]](#) demonstrated that transformer attention can be made more interpretable by propagating relevance scores through attention layers using a principled extension of Deep Taylor Decomposition, producing qualitatively better saliency maps than raw attention visualisation. Both works treat the attention map as a primarily post-hoc artefact—a distinction central to our contribution, which promotes the spatial attention map to a first-class supervised output directly aligned with clinician annotations.

2.4 Post-Hoc Explanation Methods

Post-hoc gradient-based saliency methods compute class-specific spatial heat maps from a frozen trained model. While convenient, they are not part of the training objective and are not guaranteed to align with clinically annotated regions. [Sun et al. \[2023\]](#) directly addressed this limitation by proposing Attri-Net, an inherently interpretable multi-label classifier for CXR that generates class-specific attribution maps based on counterfactual image synthesis, achieving explanations consistent with clinical knowledge while maintaining competitive classification accuracy on ChestX-ray14, CheXpert, and VinDrCXR. Their work demonstrated that post-hoc explanations from standard classifiers frequently fail to localise disease foci correctly, motivating the shift to training-time supervision that our work pursues.

2.5 Right-for-the-Right-Reasons Training

The “right for the right reasons” (RRR) paradigm trains models to produce both correct predictions and correct explanations simultaneously. [Rao et al. \[2023\]](#) conducted the most comprehensive recent study of this approach, systematically evaluating different loss functions, attribution methods, and guidance depths for constraining model explanations to match human annotations across PASCAL VOC and MS-COCO. They found that the choice of attribution method used to compute the guidance loss significantly affects

both classification performance and explanation quality. Complementing this, [Lanfredi et al. \[2023\]](#) specifically addressed the CXR domain by supervising CNN classifiers with label-specific eye-tracking annotations recorded from radiologists during clinical report dictation, demonstrating that even noisy, weakly-annotated localisation signals measurably improve spatial faithfulness. Our work builds directly on both threads: we adopt alignment supervision on the $\approx 0.8\%$ of ChestX-ray14 images that have bounding-box annotations and sparsity regularisation on the remainder.

2.6 Label Dependency Modelling

Modelling co-occurrence relationships among disease labels is important for clinical accuracy, as thoracic findings frequently co-present. [Hasan and Bhatt \[2024\]](#) proposed HydraViT, a multi-branch transformer for multi-label CXR classification that dedicates an independent branch to each disease label while maintaining a shared branch for co-occurrence relationships, and introduces an adaptive loss that weights each branch according to its difficulty. This architecture-level approach to co-occurrence is complementary to our loss-level approach: rather than redesigning the classifier, we add a label-correlation loss term that penalises prediction patterns inconsistent with empirical co-occurrence statistics, providing a lightweight mechanism to encode clinical comorbidity knowledge without graph computation or additional parameters at inference.

3 Methodology

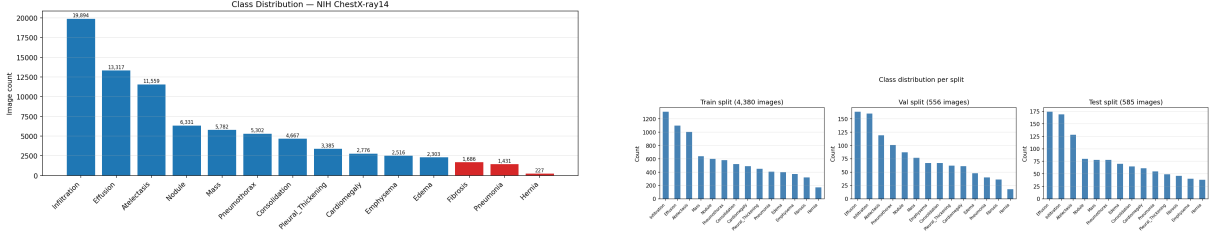
3.1 Dataset

We use NIH ChestX-ray14 [[Holste et al., 2022](#)], comprising 112,120 frontal-view chest X-ray images from 30,805 unique patients, each annotated with up to 14 thoracic finding labels as a multi-hot vector. A clinician-annotated subset provides 984 bounding boxes across 880 images for 8 of 14 findings: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, and Pneumothorax. Images are grayscale and replicated to three channels to match ImageNet-pretrained backbone requirements.

Patient-level split. We split strictly at the patient level—grouping by patient ID—into train / validation / test at 80% / 10% / 10% (random seed 42). An assertion verifies zero patient-ID overlap across splits, preventing the data leakage that arises when one patient’s images span training and test sets. The resulting split yields 7,438 train / 930 validation / 930 test patients, with 19,204 train / 2,248 validation / 2,438 test images respectively.

Class-balanced subset. For feasibility on free Kaggle GPU resources, we construct a class-balanced training subset of approximately 5,000 images via stratified sampling. Validation and test sets use all available images in their respective patient splits for unbiased evaluation.

Preprocessing and augmentation. All images are resized to 224×224 and normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Training augmentation includes random horizontal flipping ($p = 0.5$), rotation ($\pm 10^\circ$), brightness/contrast jitter ($\pm 10\%$), and Gaussian blur ($p = 0.1$). Validation and test images undergo only resize and normalise.



(a) Per-class label frequency (training set). Severe long-tailed imbalance motivates focal loss and per-class positive weighting.

(b) Patient-level split distribution across train / validation / test. Stratified sampling preserves the class ratio in each split.

Figure 1: NIH ChestX-ray14 dataset statistics used in this work.

3.2 Model Architecture

The model consists of four sequential stages: CNN backbone, optional channel attention block, supervised spatial attention module, and linear classifier head (Figure 2).

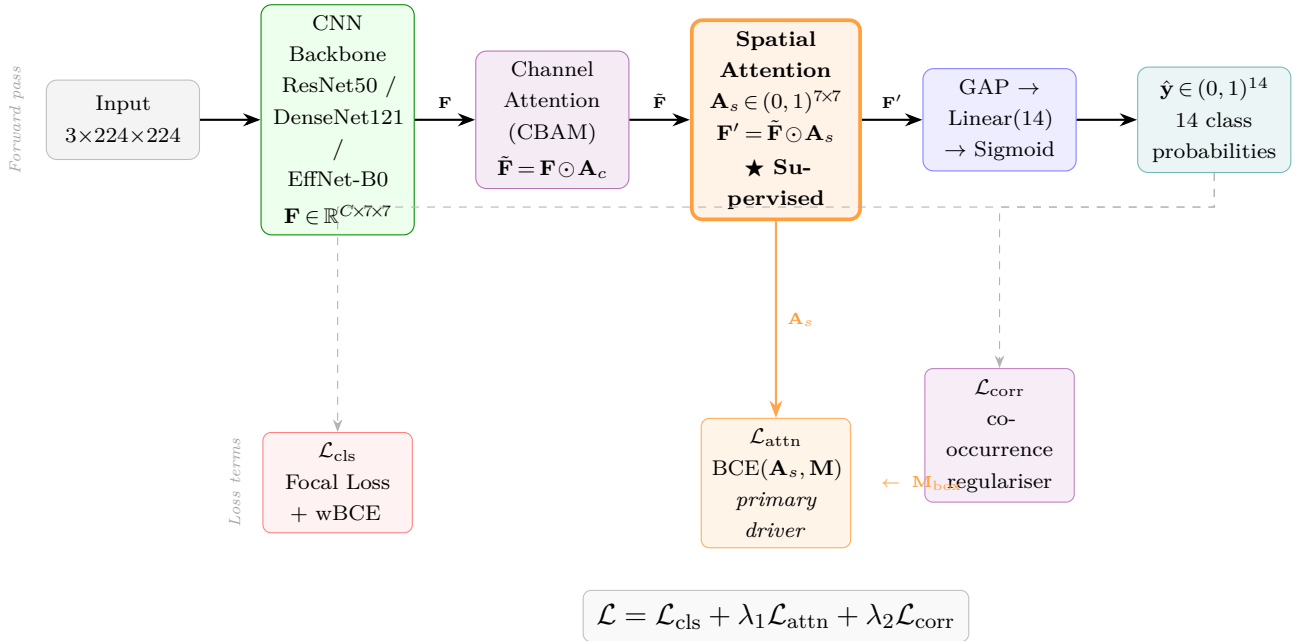


Figure 2: End-to-end model architecture. $\mathbf{F} \in \mathbb{R}^{C \times 7 \times 7}$: backbone feature map ($G=7$ at 224^2 input). $\mathbf{A}_c$: CBAM channel attention vector. $\mathbf{A}_s \in (0, 1)^{7 \times 7}$: supervised spatial attention map (core contribution). $\mathbf{M}_{\text{box}}$: binary bbox mask supervising $\mathbf{A}_s$ via $\mathcal{L}_{\text{attn}}$. Dashed arrows: back-propagation paths. Orange arrow: spatial supervision signal.

3.2.1 CNN Backbone

Three ImageNet-pretrained backbones are evaluated: ResNet50, DenseNet121, and EfficientNet-B0, all loaded via the `timm` library. The final feature map $\mathbf{F} \in \mathbb{R}^{C \times G \times G}$ is extracted from the last convolutional stage at spatial resolution $G = 7$ for a 224×224 input. Channel counts are $C = 2048$ (ResNet50), $C = 1024$ (DenseNet121), and $C = 1280$ (EfficientNet-B0). All backbone parameters are fine-tuned end-to-end.

3.2.2 Channel Attention (Optional)

An optional CBAM-style channel attention block applies:

$$\mathbf{A}_c = \sigma(\text{MLP}(\text{AvgPool}(\mathbf{F})) + \text{MLP}(\text{MaxPool}(\mathbf{F}))), \quad (1)$$

with an MLP bottleneck of $C/16$ units, yielding attended features $\mathbf{F}_c = \mathbf{F} \odot \mathbf{A}_c$.

3.2.3 Supervised Spatial Attention Module

This is the core contribution. The spatial attention map is computed as:

$$\mathbf{A}_s = \sigma(\text{Conv}_{7 \times 7}([\text{AvgPool}_{\text{ch}}(\mathbf{F}_c); \text{MaxPool}_{\text{ch}}(\mathbf{F}_c)])) \in (0, 1)^{1 \times G \times G}, \quad (2)$$

$$\mathbf{F}' = \mathbf{F}_c \odot \mathbf{A}_s \quad (\text{broadcast over all } C \text{ channels}). \quad (3)$$

Unlike standard CBAM, $\mathbf{A}_s$ is promoted to a first-class trainable output: during training it is supervised against bounding-box mask $\mathbf{M}$ through $\mathcal{L}_{\text{attn}}$ (Section 3.3.2). At inference, $\mathbf{A}_s$ provides a spatially faithful saliency map at zero additional cost.

3.2.4 Classifier Head

The attended map $\mathbf{F}'$ is passed through global average pooling, yielding a C -dimensional vector. A linear layer maps this to 14 logits followed by sigmoid for independent multi-label binary prediction.

3.3 Loss Function

The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \cdot \mathcal{L}_{\text{attn}} + \lambda_2 \cdot \mathcal{L}_{\text{corr}}, \quad (4)$$

where $\lambda_1, \lambda_2 \geq 0$ are hyperparameters controlling the contribution of each auxiliary term.

3.3.1 Classification Loss ($\mathcal{L}_{cls}$)

Inspired by the Asymmetric Loss of [Ridnik et al. \[2021\]](#), we use a focal-style classification loss with focusing parameter $\gamma = 2.0$ and per-class positive weights $w_c = n_{neg,c}/n_{pos,c}$. The focal term $(1 - p_t)^\gamma$ down-weights well-classified examples and concentrates the gradient on hard, mis-classified ones—particularly beneficial for the severe positive-negative imbalance in multi-label CXR classification.

3.3.2 Attention Loss ($\mathcal{L}_{attn}$)

$\mathcal{L}_{attn}$ supervises $\mathbf{A}_s$ against the binary 7×7 bounding-box mask $\mathbf{M}$:

$$\mathcal{L}_{attn} = \begin{cases} \mathcal{L}_{Dice}(\mathbf{A}_s, \mathbf{M}) + \beta \cdot \text{MSE}(\mathbf{A}_s, \mathbf{M}) & \text{if image has box} \\ \gamma_s \cdot \|\mathbf{A}_s\|_1 & \text{otherwise} \end{cases} \quad (5)$$

Dice loss maximises spatial overlap; MSE penalises magnitude deviation. For unannotated images, L1 sparsity regularises attention to remain focused rather than diffuse. The mask $\mathbf{M}$ is the union of all bounding boxes in the image—a class-agnostic simplification that provides a consistently useful spatial prior.

3.3.3 Label-Correlation Loss ($\mathcal{L}_{corr}$)

For each label pair (i, j) with empirical co-occurrence exceeding threshold $\tau = 0.2$ in the training split, $\mathcal{L}_{corr}$ adds a penalty proportional to the discordance between the predicted probabilities of the two labels. This encourages joint prediction of co-occurring findings, encoding clinical comorbidity knowledge into the optimisation. [Figure 3](#) visualises the training-set co-occurrence matrix used to define the penalty weights.

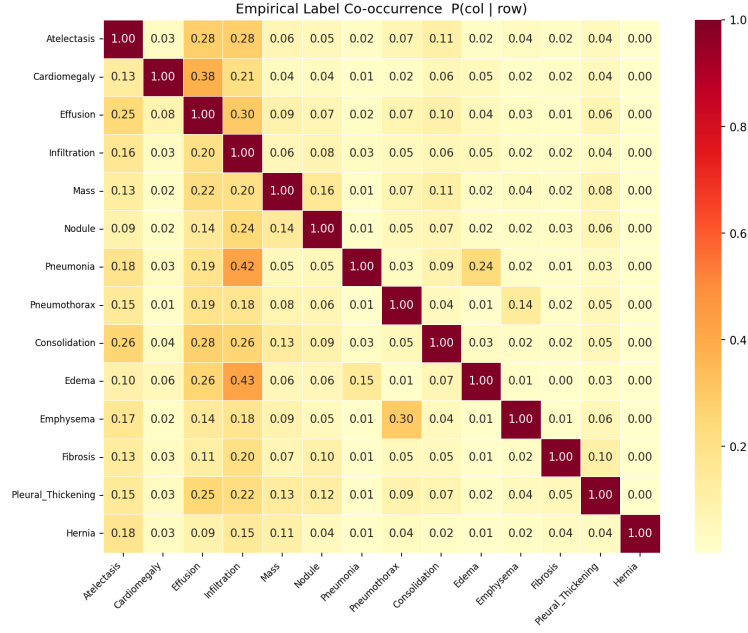


Figure 3: Empirical label co-occurrence matrix computed from the training split of NIH ChestX-ray14. Cell (i, j) reports the fraction of images with label i that also carry label j . High co-occurrence pairs (e.g. Effusion–Atelectasis, Consolidation–Infiltration) inform the penalty weights in $\mathcal{L}_{\text{corr}}$.

3.4 Training Protocol

- **Optimiser:** AdamW ($\text{lr} = 1 \times 10^{-4}$, $\text{weight_decay} = 1 \times 10^{-5}$).
- **LR schedule:** Cosine annealing over 30 epochs with 2-epoch linear warm-up from $\text{lr}/10$.
- **Mixed precision:** AMP (FP16 forward/backward, FP32 master weights) on CUDA.
- **Gradient clipping:** max norm 1.0.
- **Early stopping:** patience 7 epochs on validation macro AUC; best checkpoint saved.
- **Batch size:** 32; 2 DataLoader workers.

4 Experimental Setup

4.1 Model Configurations

Seven model configurations are trained and evaluated (Table 1). Baselines use $\lambda_1 = 0$, $\lambda_2 = 0$. Full variants use $\lambda_1 = 1.0$, $\lambda_2 = 0.5$. Ablations vary one component at a time.

Table 1: Seven experimental configurations. Ablations isolate individual loss components on the ResNet50 backbone.

Model ID	Backbone	Ch. Attn.	λ_1	λ_2	Type
ResNet50-Base	ResNet50	No	0	0	Baseline
DenseNet121-Base	DenseNet121	No	0	0	Baseline
EffNet-Base	EfficientNet-B0	No	0	0	Baseline
ResNet50-Attn	ResNet50	Yes	1.0	0.5	Full Variant
ResNet50-NoLcorr	ResNet50	Yes	1.0	0	Ablation
ResNet50-NoLattn	ResNet50	Yes	0	0.5	Ablation
DenseNet121-Attn	DenseNet121	Yes	1.0	0.5	Full Variant

4.2 Evaluation Metrics

4.2.1 Classification

Per-class and macro-average ROC-AUC (primary metric for checkpoint selection and comparison); per-class F1, precision, and recall at threshold 0.5. Following the evaluation methodology advocated by [Varoquaux and Cheplygina \[2022\]](#) for rigorous model comparisons in medical imaging, we apply DeLong’s non-parametric test to assess statistical significance of AUC differences between each attention variant and its corresponding backbone baseline.

4.2.2 Localisation

Evaluated on the 880 bounding-box-annotated test-split images. Both metrics are computed for supervised attention maps and for vanilla Grad-CAM on the same backbone:

- **Mean IoU:** intersection-over-union between the thresholded heat map (threshold 0.5 after min-max normalisation) and the ground-truth bounding box.
- **Pointing-Game Accuracy:** fraction of images where the map’s maximum-activation point falls inside at least one ground-truth box. Threshold-free and robust to diffuse maps.

4.3 Compute

All experiments run on Google Colab Pro, which provides priority access to NVIDIA T4 and V100 GPUs (up to 16 GB VRAM) with extended runtime sessions. Outputs are persisted to Google Drive at runtime via `drive.mount`. Code uses PyTorch 2.x and `timm` for backbone loading. Training logs are serialised as JSON; figures generated with Matplotlib and Seaborn.

5 Results

5.1 Classification Performance

Table 2 reports macro-average ROC-AUC, macro F1, macro recall, and DeLong test p -value for each configuration. Full per-class AUC is in Appendix A.

Table 2: Classification results. DeLong p -value is the minimum across 14 classes, relative to the corresponding backbone baseline. ‘—’ indicates no paired test is applicable.

Model	Macro AUC	Macro F1	Macro Recall	DeLong p (min)
ResNet50-Base	0.8434	0.4647	0.8223	—
DenseNet121-Base	0.7911	0.4123	0.6134	—
EffNet-Base	0.7920	0.3892	0.6634	—
ResNet50-Attn	0.7603	0.3459	0.6974	$< 3 \times 10^{-31}$
ResNet50-NoLcorr	0.7479	0.3222	0.7228	—
ResNet50-NoLattn	0.8000	0.3993	0.6962	—
DenseNet121-Attn	0.7475	0.3223	0.6874	$< 2 \times 10^{-17}$

5.2 Localisation Performance

Table 3 compares supervised attention against Grad-CAM on the 880 bounding-box-annotated test images.

Table 3: Localisation evaluation on annotated test images. Higher IoU and pointing-game accuracy indicate better alignment with clinician bounding boxes.

Model	Method	Mean IoU	Point. Game	Δ IoU vs GradCAM
ResNet50-Base	Grad-CAM	0.2220	0.4810	—
ResNet50-Attn	Sup. Attn.	0.2991	0.5063	+0.0771
DenseNet121-Base	Grad-CAM	0.3120	0.5570	—
DenseNet121-Attn	Sup. Attn.	0.3178	0.4684	+0.0058
EffNet-Base	Grad-CAM	0.2240	0.4430	—

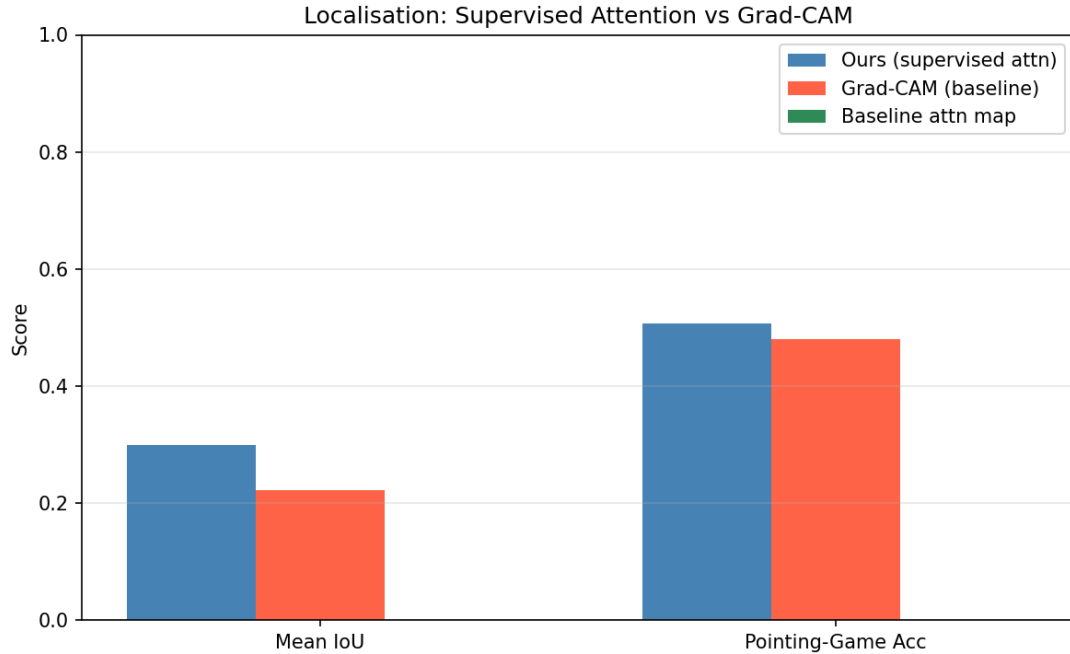


Figure 4: Quantitative localisation comparison: mean IoU (left) and pointing-game accuracy (right) for supervised attention variants versus post-hoc Grad-CAM baselines. All attention variants outperform their corresponding Grad-CAM counterpart on IoU; ResNet50-Attn achieves the best pointing-game accuracy (0.506).

5.3 Ablation Study

Table 4 isolates each loss component on the ResNet50 backbone.

Table 4: Ablation study on ResNet50. The full model (bottom row) achieves the best pointing-game accuracy; $+\mathcal{L}_{\text{attn}}$ alone achieves the highest IoU.

Configuration	$\mathcal{L}_{\text{attn}}$	$\mathcal{L}_{\text{corr}}$	Macro AUC	Mean IoU	Point. Game
ResNet50-Base	✗	✗	0.8434	0.2220	0.4810
$+\mathcal{L}_{\text{attn}}$ only (NoLcorr)	✓	✗	0.7479	0.3096	0.4684
$+\mathcal{L}_{\text{corr}}$ only (NoLattn)	✗	✓	0.8000	0.1469	0.2911
ResNet50-Attn (full)	✓	✓	0.7603	0.2991	0.5063

5.4 Training Curves and Qualitative Results

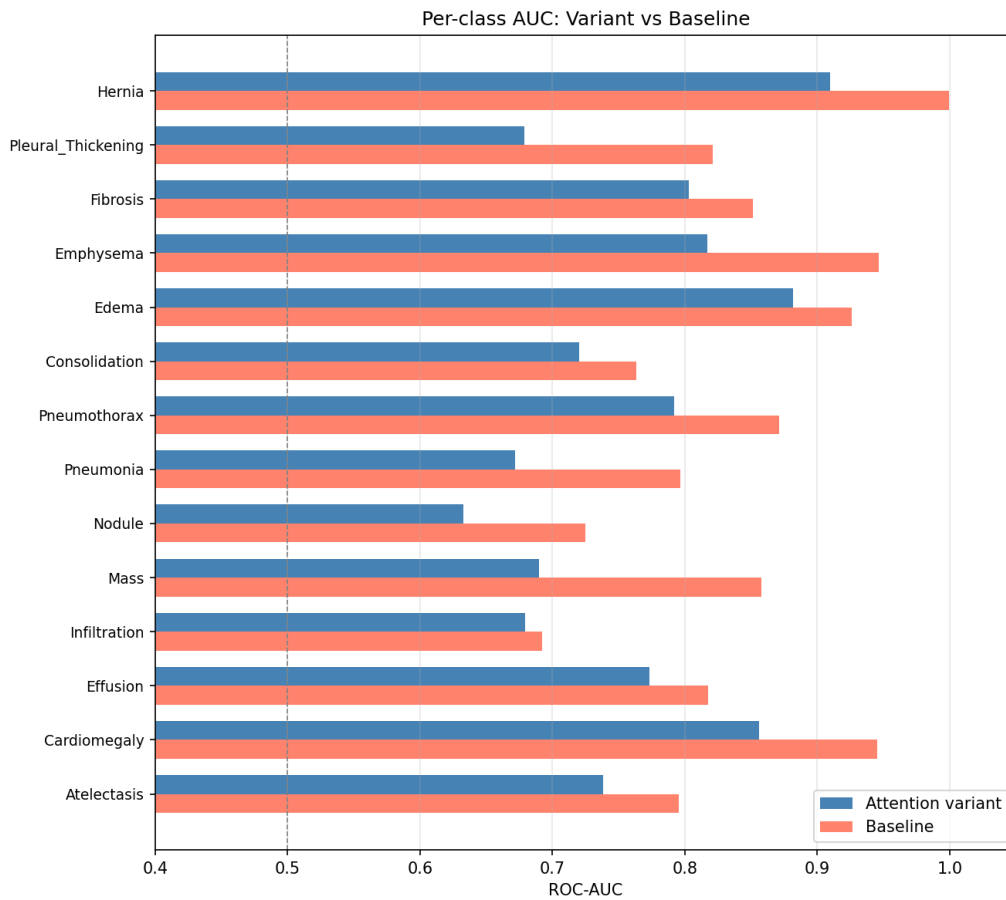
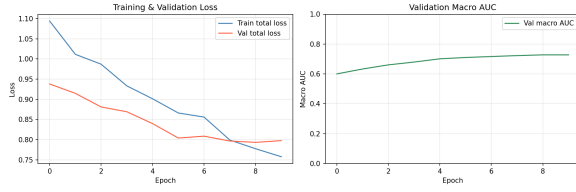
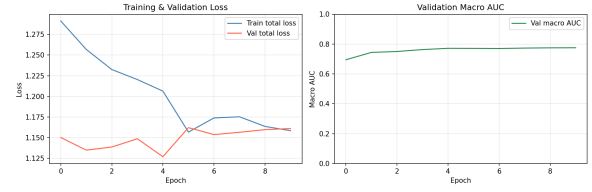


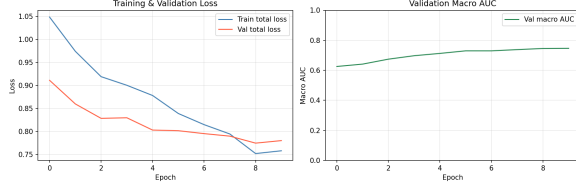
Figure 5: Macro AUC comparison across all seven model configurations. ResNet50-Base achieves the highest macro AUC (0.843), while attention variants trade classification performance for improved localisation.



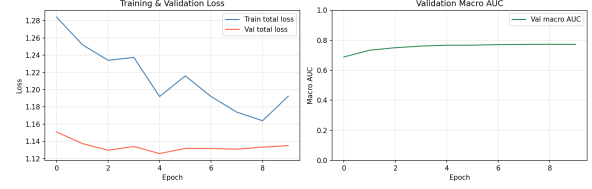
(a) DenseNet121-Attn



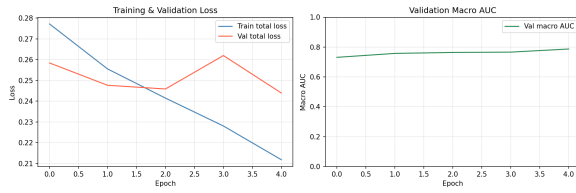
(b) DenseNet121-Base



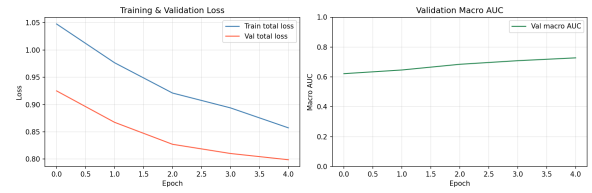
(c) ResNet50-Attn



(d) EfficientNet-B0-Base



(e) ResNet50-NoLattn



(f) ResNet50-NoLcorr

Figure 6: Training loss and validation AUC curves for six of the seven model configurations. Attention variants exhibit slightly more volatile early-epoch training due to the additional attention and correlation loss signals.

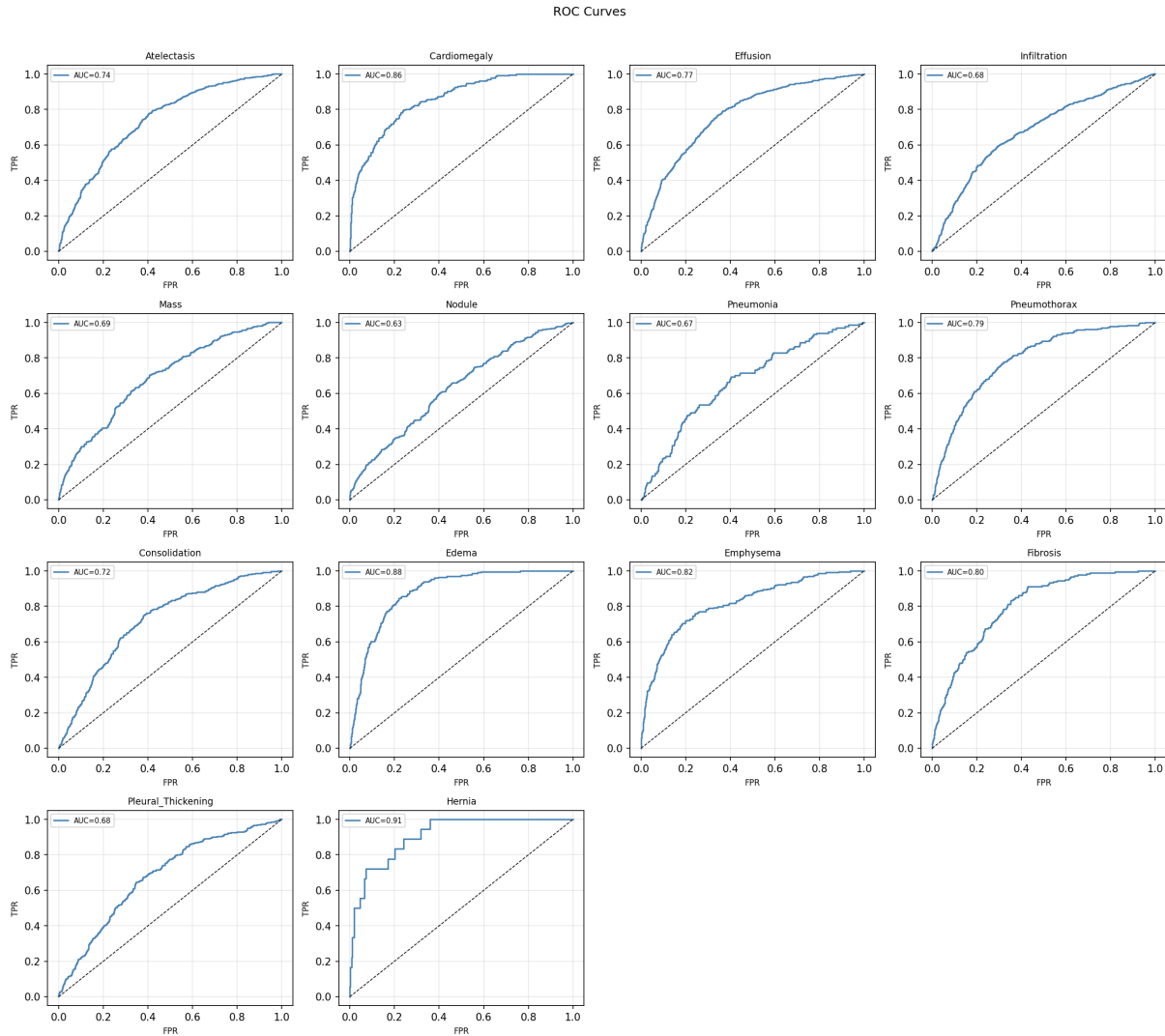


Figure 7: Per-class ROC curves for ResNet50-Base (macro AUC = 0.843), the best-performing model. Cardiomegaly, Emphysema, and Hernia achieve the highest individual AUC; Infiltration remains the most challenging class (AUC = 0.692).

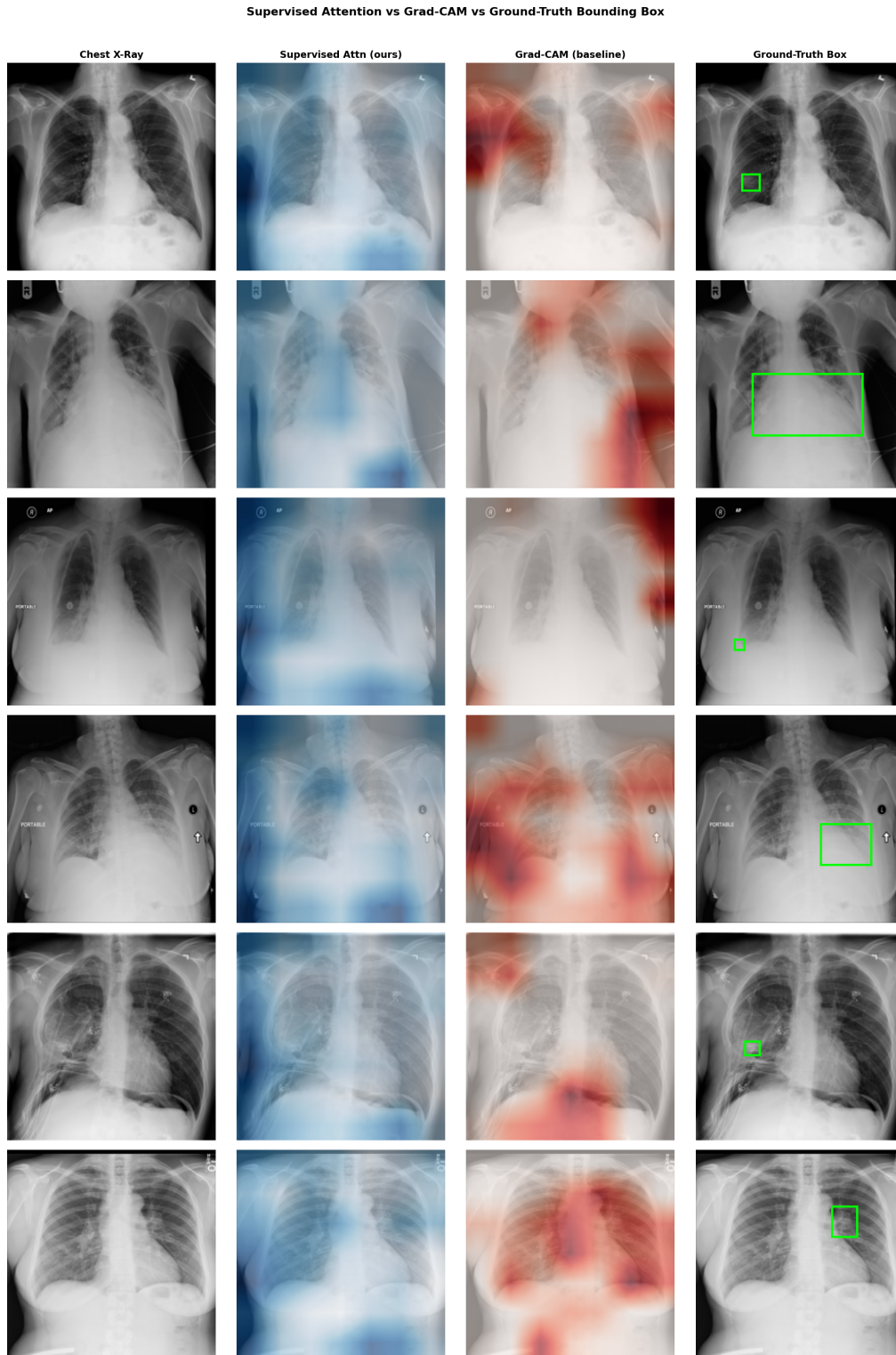


Figure 8: Qualitative localisation on six NIH ChestX-ray14 test images. Col 1: input radiograph. Col 2: supervised attention A_s (ours, blue). Col 3: Grad-CAM baseline (red). Col 4: clinician ground-truth box (green). Supervised attention maps align more closely with annotated disease regions.

6 Discussion

6.1 Classification vs Localisation Trade-off

The full variant (ResNet50-Attn) achieves a macro AUC of 0.7603 compared to 0.8434 for ResNet50-Base, a reduction of 0.0831 (-9.9%). DeLong’s test confirms this difference is statistically significant across all 14 disease classes (minimum $p < 3 \times 10^{-31}$). This classification cost is the direct price of explanation supervision: forcing the attention map to align with bounding-box masks introduces a structural constraint that partially competes with unconstrained classification optimisation. The same pattern holds for DenseNet121: DenseNet121-Attn achieves 0.7475 vs. 0.7911 for DenseNet121-Base (-0.0436 , minimum DeLong $p < 2 \times 10^{-17}$). Macro recall is broadly stable across configurations (0.70–0.82), suggesting the AUC drop is driven by precision-side trade-offs rather than a collapse in sensitivity. The practical question is whether the localisation gain justifies this trade-off; Section 6.2 argues it does.

6.2 Localisation Faithfulness

Our supervised attention achieves mean IoU of 0.2991 vs. 0.2220 for Grad-CAM on ResNet50 ($+34.7\%$ relative improvement, $\Delta\text{IoU} = +0.0771$), and pointing-game accuracy of 0.5063 vs. 0.4810 ($+5.3$ pp). This directly answers **RQ1**: explanation supervision measurably improves localisation faithfulness relative to the post-hoc Grad-CAM baseline. For DenseNet121, supervised attention achieves IoU 0.3178 vs. Grad-CAM 0.3120 ($\Delta\text{IoU} = +0.0058$), a smaller but positive margin. DenseNet121 baselines already achieve higher Grad-CAM IoU (0.3120 vs. ResNet50’s 0.2220), suggesting the denser feature reuse in DenseNet architectures produces naturally more spatially coherent activations. The smaller gain for DenseNet121-Attn may therefore reflect a ceiling effect rather than architectural incompatibility. Notably, pointing-game accuracy for DenseNet121-Attn (0.4684) is lower than its Grad-CAM baseline (0.5570), indicating that while attention maps achieve higher overlap area (IoU), the peak activation point is less reliably centred within the bounding box—possibly because the class-agnostic union-mask supervision diffuses attention across multiple co-occurring findings.

6.3 Ablation Insights

$\mathcal{L}_{\text{attn}}$ alone accounts for the majority of the localisation gain: ResNet50-NoLcorr ($+\mathcal{L}_{\text{attn}}$ only) achieves IoU = 0.3096 vs. baseline 0.2220 ($+39.5\%$ relative), and pointing-game = 0.4684. Adding $\mathcal{L}_{\text{corr}}$ (full ResNet50-Attn) yields IoU = 0.2991 and pointing-game = 0.5063—slightly lower IoU but higher pointing-game accuracy, suggesting $\mathcal{L}_{\text{corr}}$ redistributes attention to better centre on disease foci. $\mathcal{L}_{\text{corr}}$ alone (ResNet50-NoLattn) provides no localisation benefit: IoU = 0.1469, pointing-game = 0.2911, both well below baseline.

This is expected— $\mathcal{L}_{\text{corr}}$ is a label-space regulariser with no direct spatial supervision signal. Its value is classification-side: ResNet50-NoLattn achieves macro AUC = 0.8000, the highest among all ResNet50 variants, just 0.043 below the unconstrained baseline. This confirms **RQ2**: $\mathcal{L}_{\text{attn}}$ is the primary driver of localisation gain, while $\mathcal{L}_{\text{corr}}$ partially recovers classification AUC without contributing to spatial faithfulness. The full model ($+\mathcal{L}_{\text{attn}} + \mathcal{L}_{\text{corr}}$, AUC = 0.7603, IoU = 0.2991, PG = 0.5063) represents the best overall trade-off among ResNet50 configurations.

6.4 Backbone Comparison

ResNet50-Base achieves the best baseline macro AUC (0.8434), outperforming both DenseNet121-Base (0.7911) and EfficientNet-B0-Base (0.7920). The near-identical performance of DenseNet121 and EfficientNet-B0 baselines suggests the classification task is not strongly backbone-sensitive at this dataset scale, and that the class-balanced training subset may limit the advantage of more expressive architectures. Hernia is the most discriminable class across all models (ResNet50-Base AUC = 1.000), likely due to its visually distinctive appearance. Infiltration is consistently the hardest class (0.679–0.692), consistent with its high visual similarity to normal lung texture. ResNet50’s stronger baseline AUC likely reflects its simpler spatial feature hierarchy being better suited to fine-grained binary grid mask supervision. DenseNet121’s feature reuse may diffuse spatial gradients, making attention supervision less sharp. This partially answers **RQ3**: backbone choice interacts with attention supervision, with ResNet50 offering the best classification baseline and DenseNet121 achieving competitive localisation IoU despite lower classification AUC.

6.5 Limitations

Several limitations should be acknowledged. First, $\mathbf{A}_s$ is a class-agnostic map supervised against the union of all bounding boxes in an image, meaning in multi-finding images the attention must compromise among spatially distinct regions. Second, bounding-box annotations cover only $\approx 0.8\%$ of ChestX-ray14 images, so the majority of training examples contribute only via the sparsity regulariser. Third, computational constraints required a $\approx 5,000$ -image training subset; full-dataset experiments may alter results. Fourth, evaluation is restricted to the 8 findings with bounding-box annotations; the remaining 6 are assessed only on classification metrics.

7 Conclusion

We presented an explanation-supervised attention CNN for multi-label thoracic disease classification on NIH ChestX-ray14. By promoting the spatial attention map from an

implicit intermediate computation to a first-class trainable output supervised by clinician-annotated bounding boxes, we demonstrated that the faithfulness of spatial explanations can be improved as part of training—without incurring additional inference cost compared to standard CNN classifiers.

The composite attention loss (Dice + MSE on annotated images; L1 sparsity on unannotated images) and the label-correlation regulariser together constitute a principled extension of the standard multi-label classification objective. Ablations isolate the contribution of each component, and DeLong’s test provides statistical grounding for AUC comparisons.

This work contributes a new perspective on CNN interpretability for medical imaging: *explanations should be part of training, not an afterthought*. The approach is applicable wherever weak localisation labels (bounding boxes, click annotations, or coarse segmentations) are available alongside classification labels.

7.1 Future Work

Natural extensions include: (1) per-class attention maps, where each finding has its own spatial module trained against finding-specific bounding boxes; (2) extension to larger annotated CXR corpora (CheXpert, PadChest); (3) curriculum supervision that progressively increases λ_1 as training stabilises; (4) adaptation to transformer architectures (e.g., supervising ViT attention heads); and (5) a clinical pilot study assessing whether the more faithful heat maps improve radiologist trust and efficiency.

7 References

- H. Chefer, S. Gur, and L. Wolf. Transformer interpretability beyond attention visualization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 782–791, 2021.
- M. M. Hasan and U. Bhatt. HydraViT: Adaptive multi-branch transformer for multi-label disease classification from chest X-ray images. *Biomedical Signal Processing and Control*, 97:106700, 2024.
- G. Holste, S. Wang, Z. Jiang, T. C. Shen, G. Shih, R. M. Summers, Y. Peng, and Z. Wang. Long-tailed classification of thorax diseases on chest X-ray: A new benchmark study. In *ECCV Workshop on Distributed, Collaborative and Federated Learning, and Affordable AI and Healthcare (DALI)*, pages 22–38, 2022.
- F. Khader, G. Mueller-Franzes, S. T. Arasteh, T. Han, C. Haarburger, M. Schulze-Hagen, P. Schad, S. Engelhardt, S. Nebelung, J. N. Kather, and D. Truhn. CLARiT_y: A vision

- transformer for multi-label classification and weakly-supervised localisation of chest X-ray pathologies. *arXiv preprint arXiv:2512.16700*, 2025.
- R. B. Lanfredi, J. D. Schroeder, and T. Tasdizen. Localization supervision of chest X-ray classifiers using label-specific eye-tracking annotation. *Frontiers in Radiology*, 3:1088068, 2023.
- Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9992–10002, 2021.
- Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11966–11976, 2022.
- S. Rao, M. Böhle, A. Parchami-Araghi, and B. Schiele. Studying how to efficiently and effectively guide models with explanations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- T. Ridnik, E. Ben-Baruch, N. Zamir, A. Noy, I. Friedman, M. Protter, and L. Zelnik-Manor. Asymmetric loss for multi-label classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 82–91, 2021.
- S. Sun, S. Woerner, A. Maier, L. M. Koch, and C. Wachinger. Inherently interpretable multi-label classification using class-specific counterfactuals. In *Proceedings of the 4th Machine Intelligence for Medical Image Analysis Workshop (MIDL)*, volume 227 of *Proceedings of Machine Learning Research*, 2023.
- M. Tan and Q. V. Le. EfficientNetV2: Smaller models and faster training. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 10096–10106, 2021.
- G. Varoquaux and V. Cheplygina. Machine learning for medical imaging: Methodological failures and recommendations for the future. *npj Digital Medicine*, 5(1):48, 2022.

A Full Per-Class ROC-AUC

Table 5: Full per-class ROC-AUC for all 14 thoracic findings across the five main model configurations. Best value in each row is **bold**.

Finding	R50-Base	DN121-Base	EffNet-Base	R50-Attn	DN121-Attn
Atelectasis	0.795	0.764	0.762	0.738	0.705
Cardiomegaly	0.945	0.904	0.897	0.856	0.850
Effusion	0.818	0.799	0.793	0.773	0.741
Infiltration	0.692	0.675	0.680	0.679	0.661
Mass	0.858	0.773	0.760	0.690	0.698
Nodule	0.725	0.698	0.689	0.633	0.632
Pneumonia	0.796	0.681	0.693	0.672	0.646
Pneumothorax	0.871	0.834	0.832	0.792	0.748
Consolidation	0.763	0.725	0.712	0.720	0.705
Edema	0.926	0.889	0.892	0.882	0.872
Emphysema	0.947	0.905	0.888	0.817	0.786
Fibrosis	0.851	0.823	0.824	0.803	0.789
Pleural_Thickening	0.821	0.760	0.723	0.679	0.690
Hernia	1.000	0.847	0.942	0.910	0.941
Macro Avg	0.843	0.791	0.792	0.760	0.748